Linear Regression with Python

Here neighbor is a real estate agent and wants some help predicting housing prices for regions in the INDIA. It would be great if i could somehow create a model for her that allows her to put in a few features of a house and returns back an estimate of what the house would sell for.

She has asked me if i could help her out with your new data science skills. me say yes, and decide that Linear Regression might be a good path to solve this problem!

My neighbor then gives you some information about a bunch of houses in regions of the India,it is all in the data set: INDIA_Housing.csv.

The data contains the following columns:

'Avg. Area Income': Avg. Income of residents of the city house is located in. 'Avg. Area House Age': Avg Age of Houses in same city 'Avg. Area Number of Rooms': Avg Number of Rooms for Houses in same city 'Avg. Area Number of Bedrooms': Avg Number of Bedrooms for Houses in same city 'Area Population': Population of city house is located in 'Price': Price that the house sold at 'Address': Address for the house

Let's get started!

Check out the data

In [1]:
```
## Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [2]:
```
INDIAhousing = pd.read_csv("INDIA_Housing.csv")

INDIAhousing.head()
```

Out[2]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701..
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482..
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

In [3]:
```
INDIAhousing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object 
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

In [4]:
```
INDIAhousing.describe()
```

Out[4]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	9.975771e+05
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1.232669e+06
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1.471210e+06
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2.469066e+06

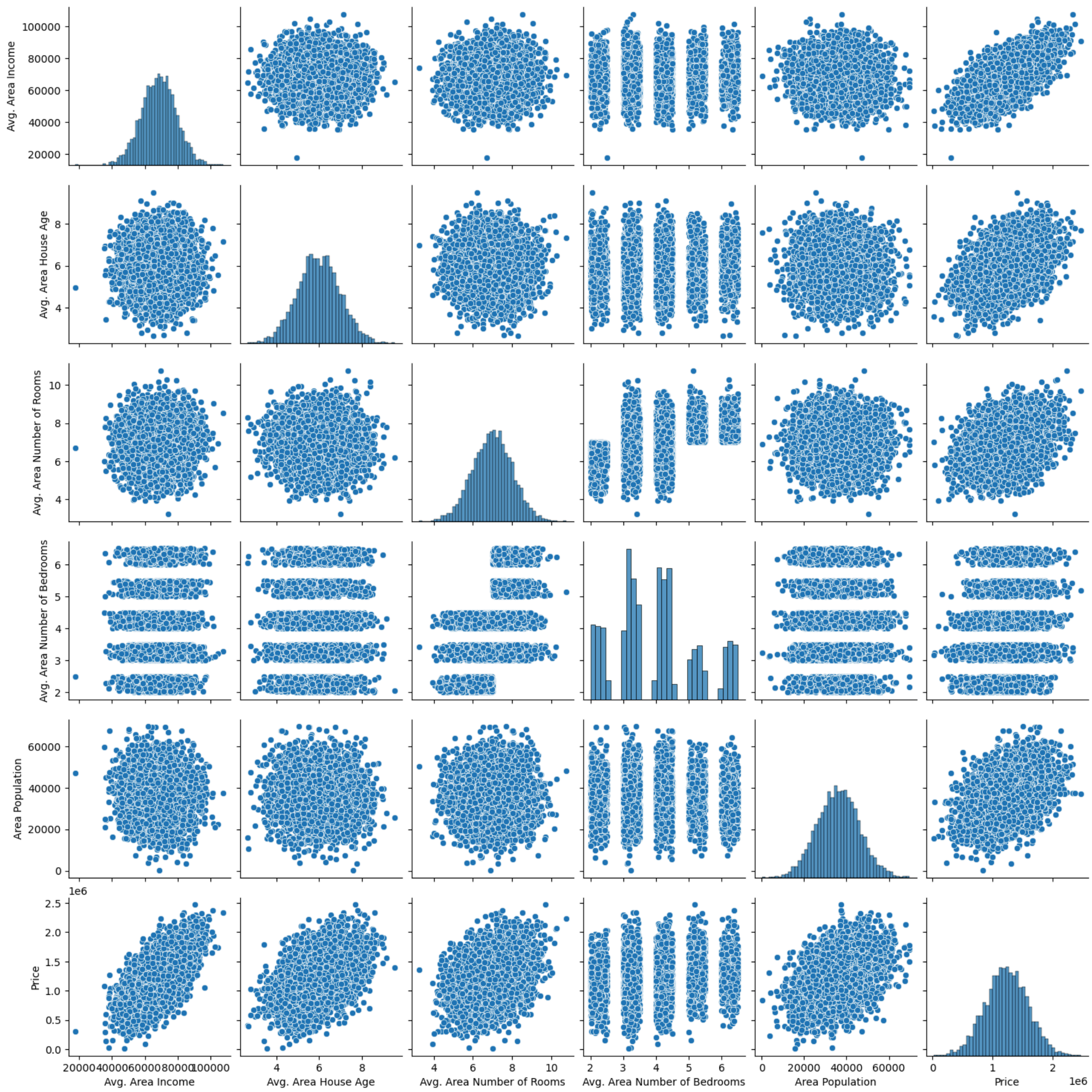
In [5]:
```
INDIAhousing.columns
```

Out[5]:
```
Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
       'Avg. Area Number of Bedrooms', 'Area Population', 'Price', 'Address'],
      dtype='object')
```

Exploratory Data Analysis for House Price Prediction

In [6]:
```
sns.pairplot(INDIAhousing)
```

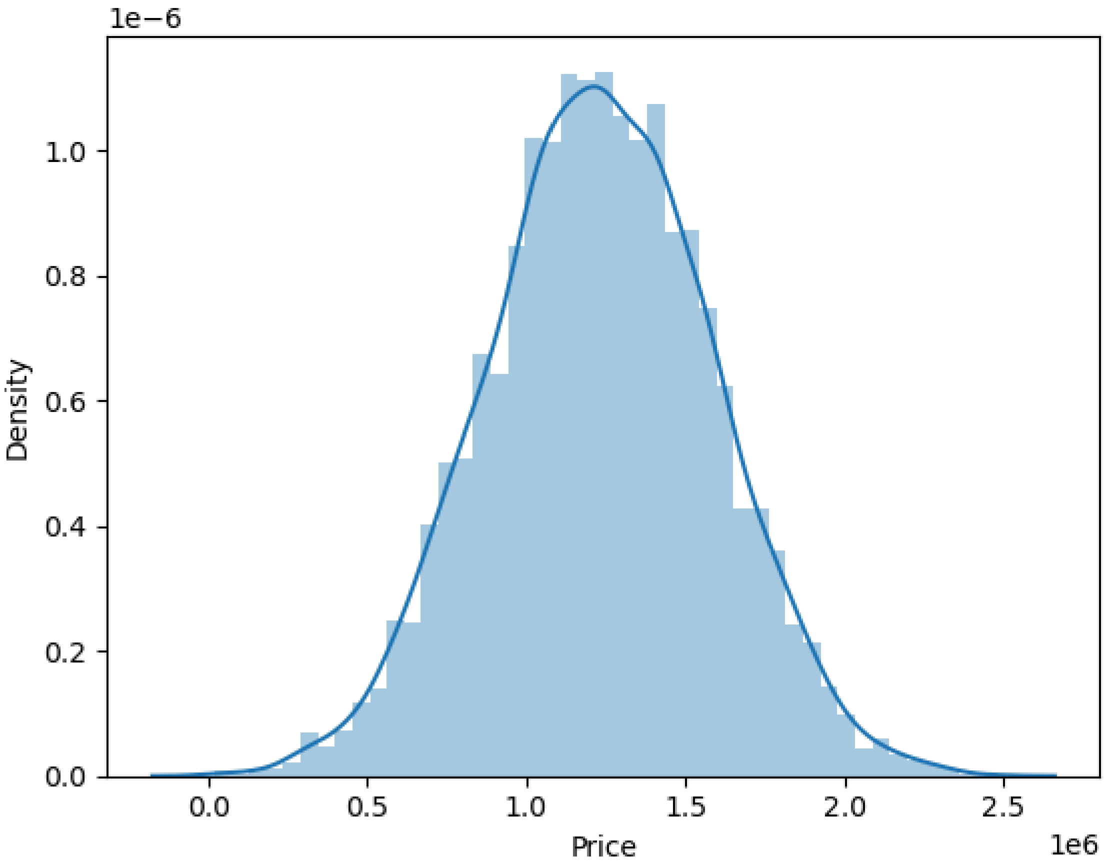
Out[6]: <seaborn.axisgrid.PairGrid at 0x21385c9ef50>



In [7]:
```
sns.distplot(INDIAhousing['Price'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\867072288.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Price'])
```

Out[7]: <Axes: xlabel='Price', ylabel='Density'>



In [8]:
```
sns.distplot(INDIAhousing['Area Population'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\2139820291.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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  sns.distplot(INDIAhousing['Area Population'])
```
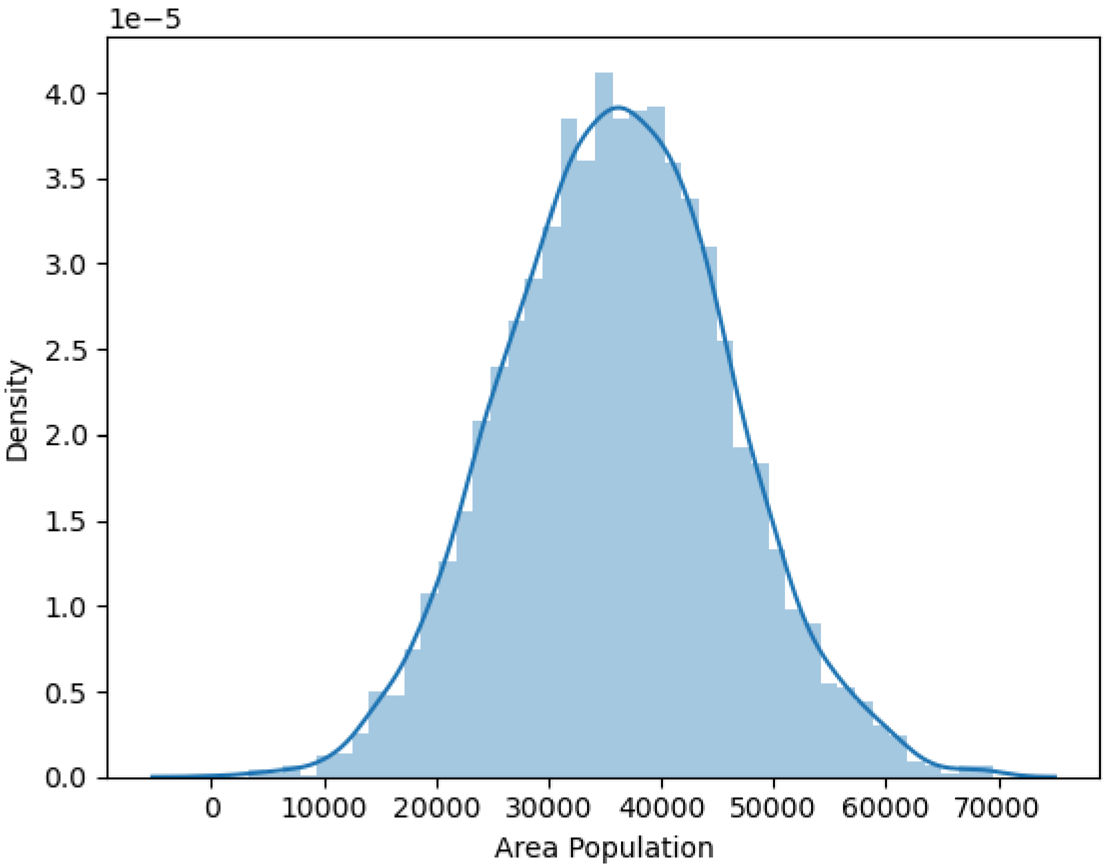

Out[8]: <Axes: xlabel='Area Population', ylabel='Density'>



In [9]:
```
sns.distplot(INDIAhousing['Avg. Area Income'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\3131757723.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Avg. Area Income'])
```
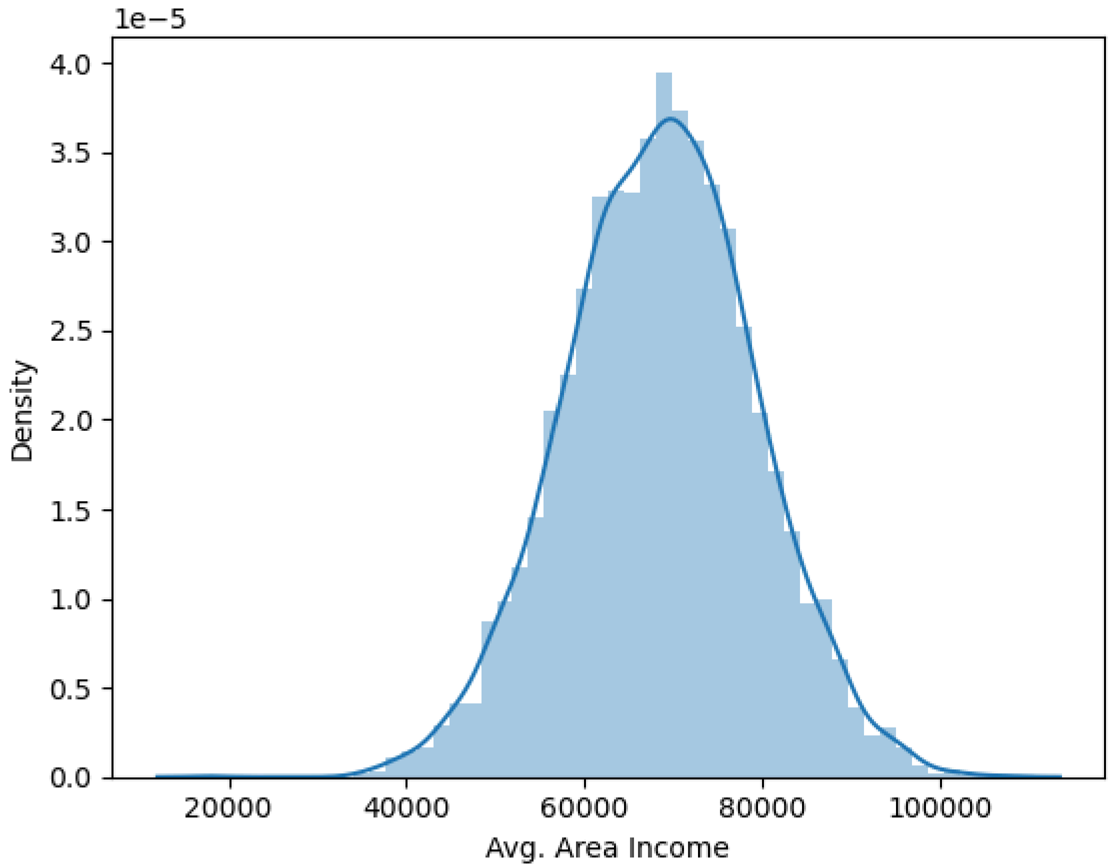

Out[9]:
```
<Axes: xlabel='Avg. Area Income', ylabel='Density'>
```



In [10]:
```
sns.distplot(INDIAhousing['Avg. Area House Age'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\1332842614.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Avg. Area House Age'])
```
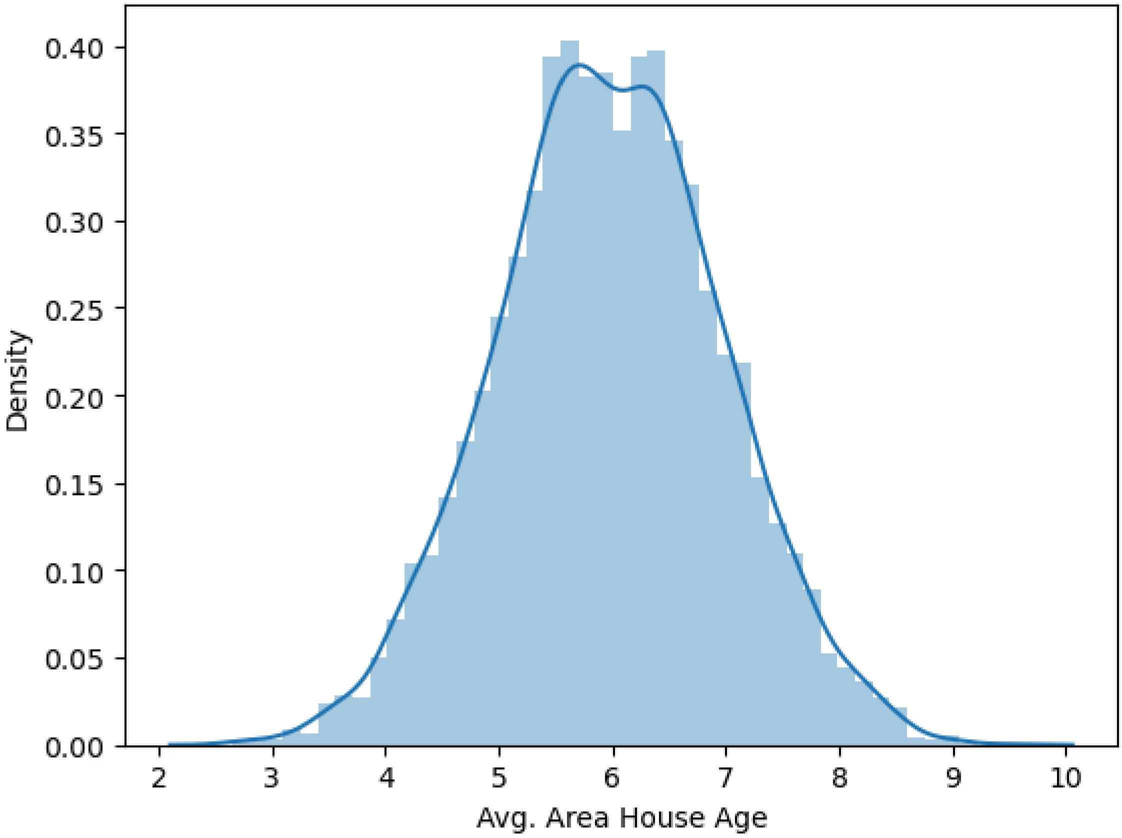

Out[10]:
```
<Axes: xlabel='Avg. Area House Age', ylabel='Density'>
```



In [11]:
```
sns.distplot(INDIAhousing['Avg. Area Number of Rooms'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\2831880010.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Avg. Area Number of Rooms'])
```
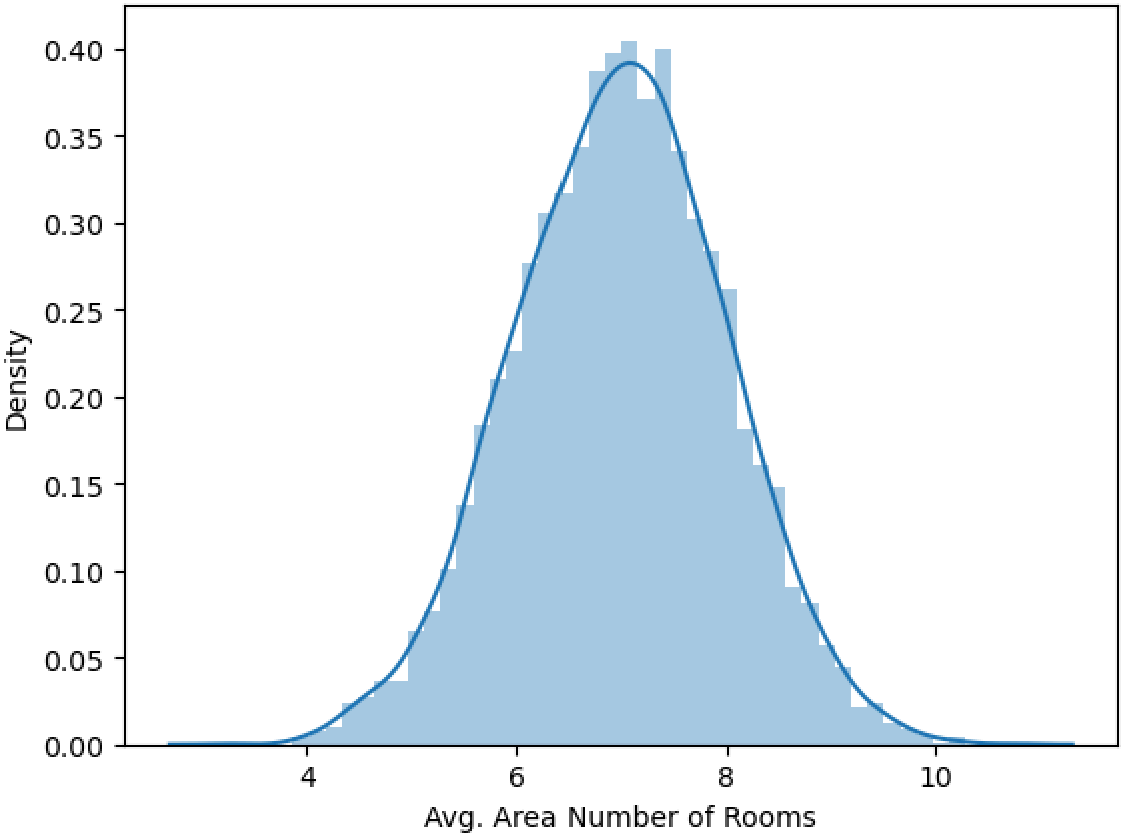

Out[11]:
```
<Axes: xlabel='Avg. Area Number of Rooms', ylabel='Density'>
```



In [12]:
```
sns.distplot(INDIAhousing['Avg. Area Number of Bedrooms'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\334197827.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Avg. Area Number of Bedrooms'])
```
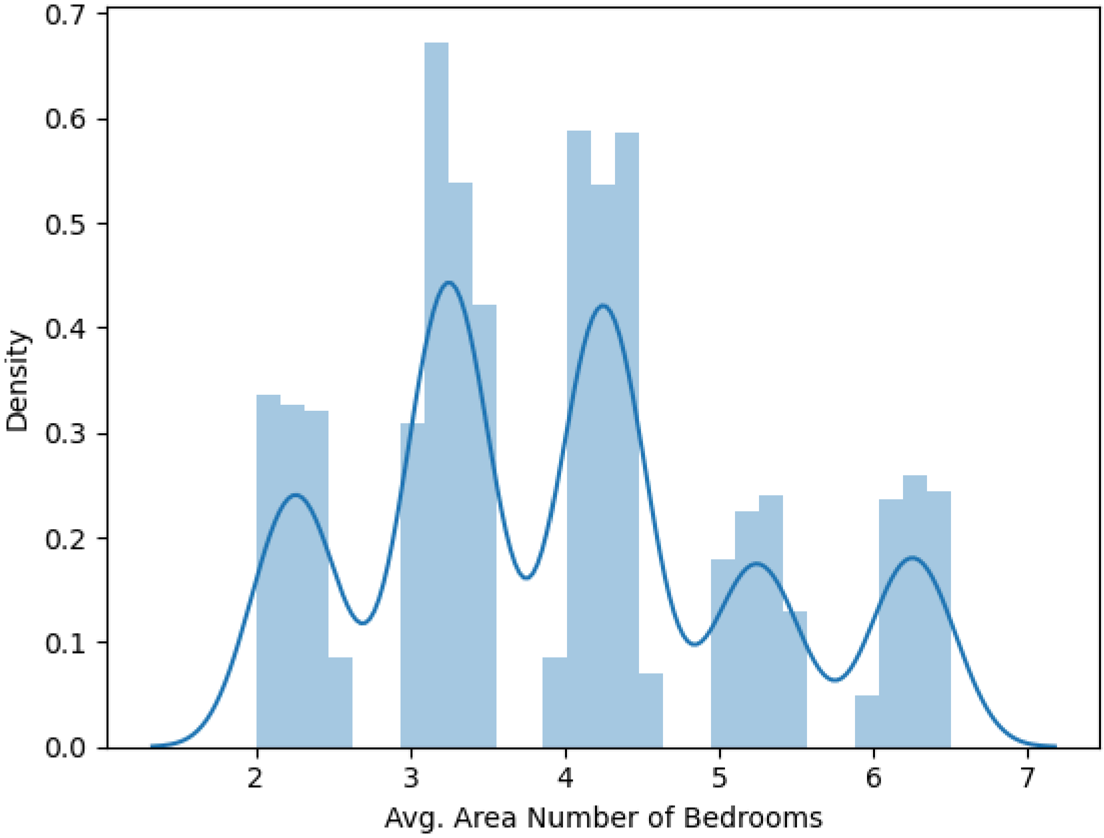

Out[12]:
```
<Axes: xlabel='Avg. Area Number of Bedrooms', ylabel='Density'>
```



In [13]:
```
sns.distplot(INDIAhousing['Area Population'])
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\2139820291.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(INDIAhousing['Area Population'])
```
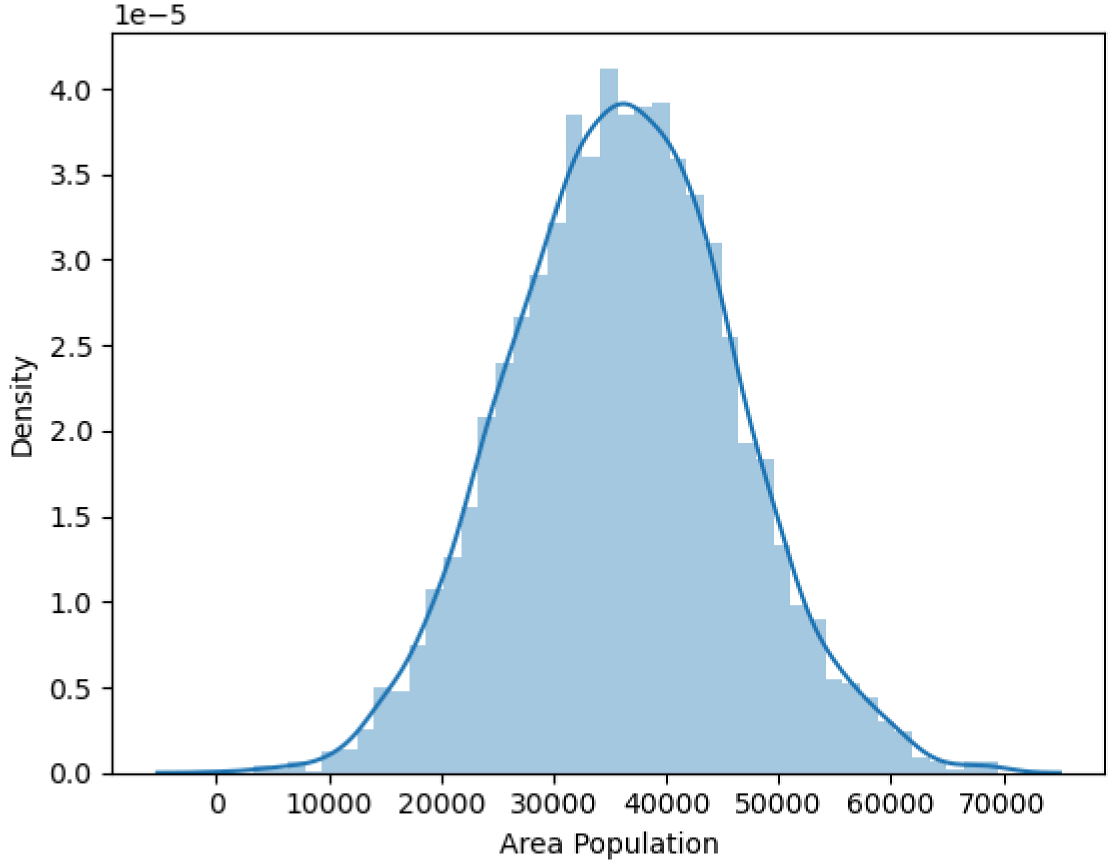
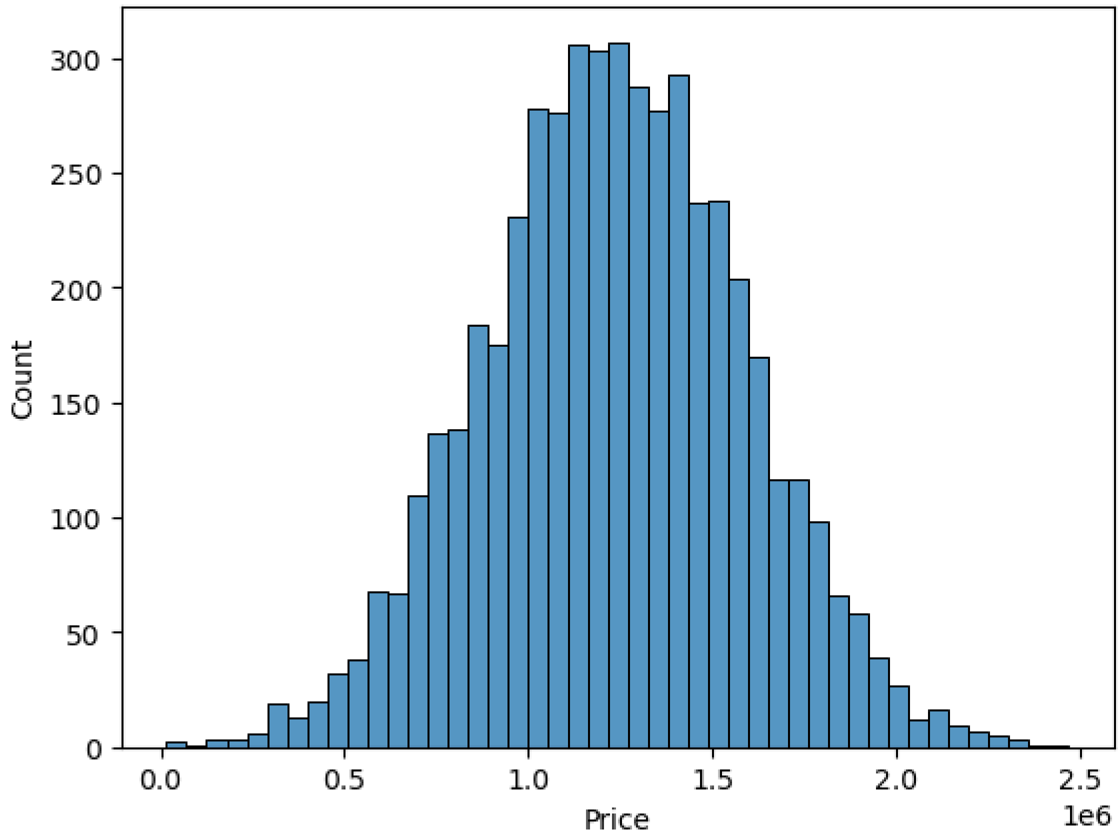

Out[13]:
```
<Axes: xlabel='Area Population', ylabel='Density'>
```



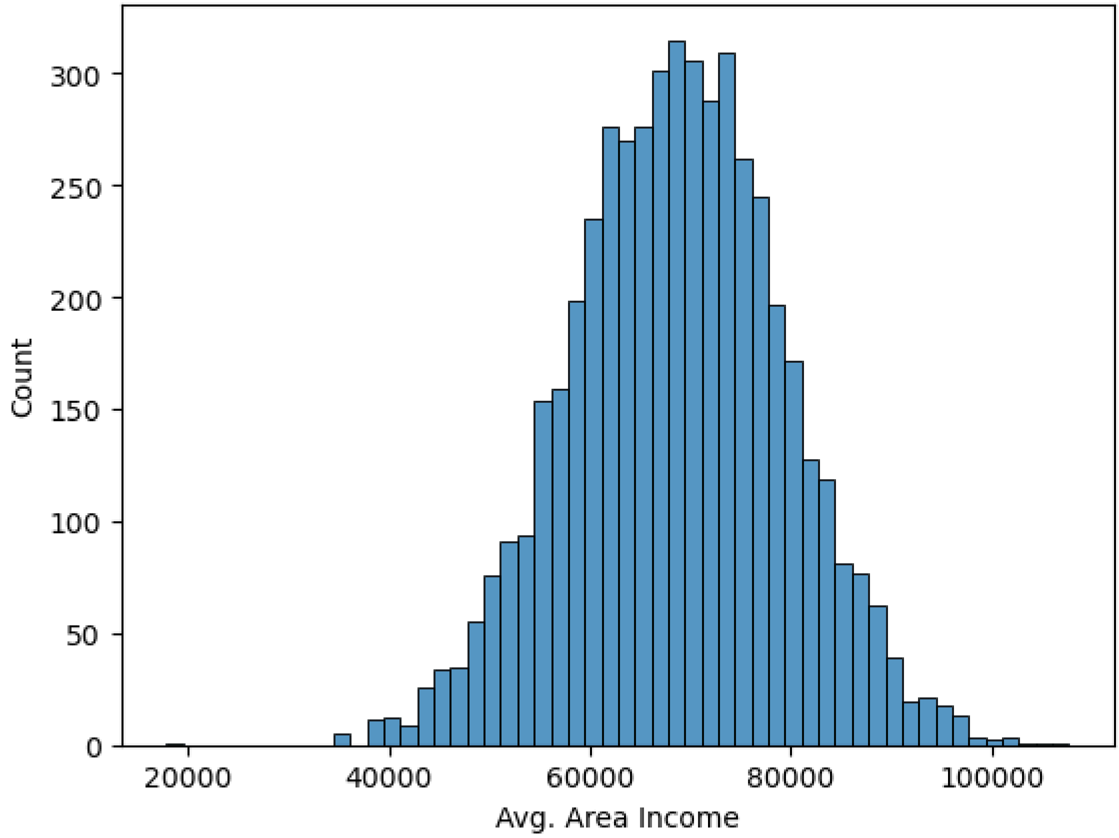
In [15]:
```
sns.histplot(INDIAhousing['Price'])
```

Out[15]: <Axes: xlabel='Price', ylabel='Count'>

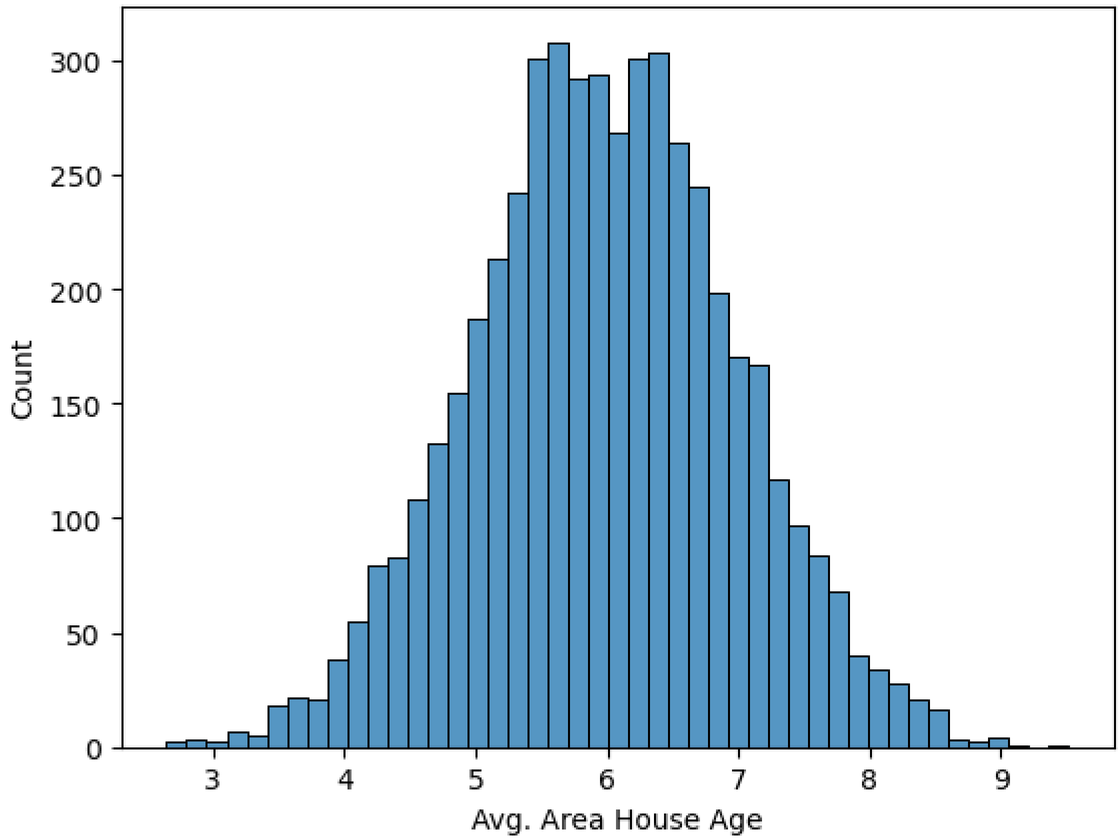


In [16]:
```
sns.histplot(INDIAhousing['Avg. Area Income'])
```

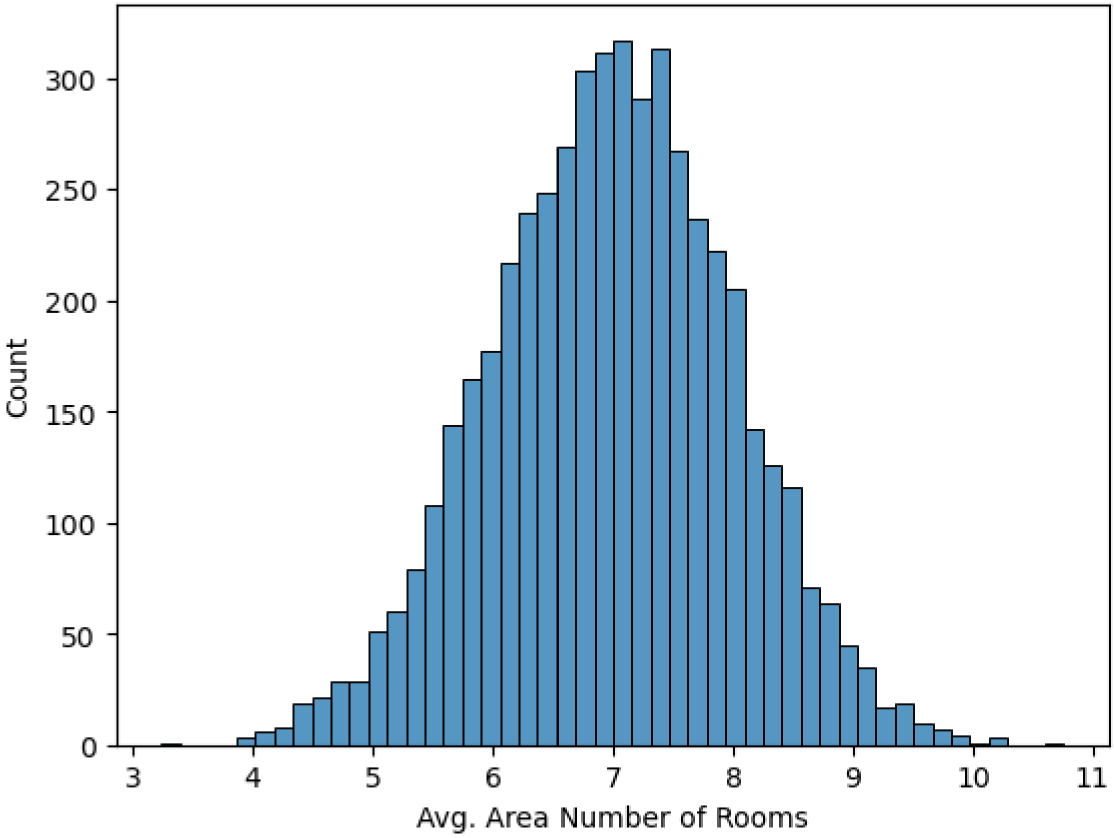
Out[16]: <Axes: xlabel='Avg. Area Income', ylabel='Count'>



In [17]:
```
sns.histplot(INDIAhousing['Avg. Area House Age'])
```
Out[17]: <Axes: xlabel='Avg. Area House Age', ylabel='Count'>

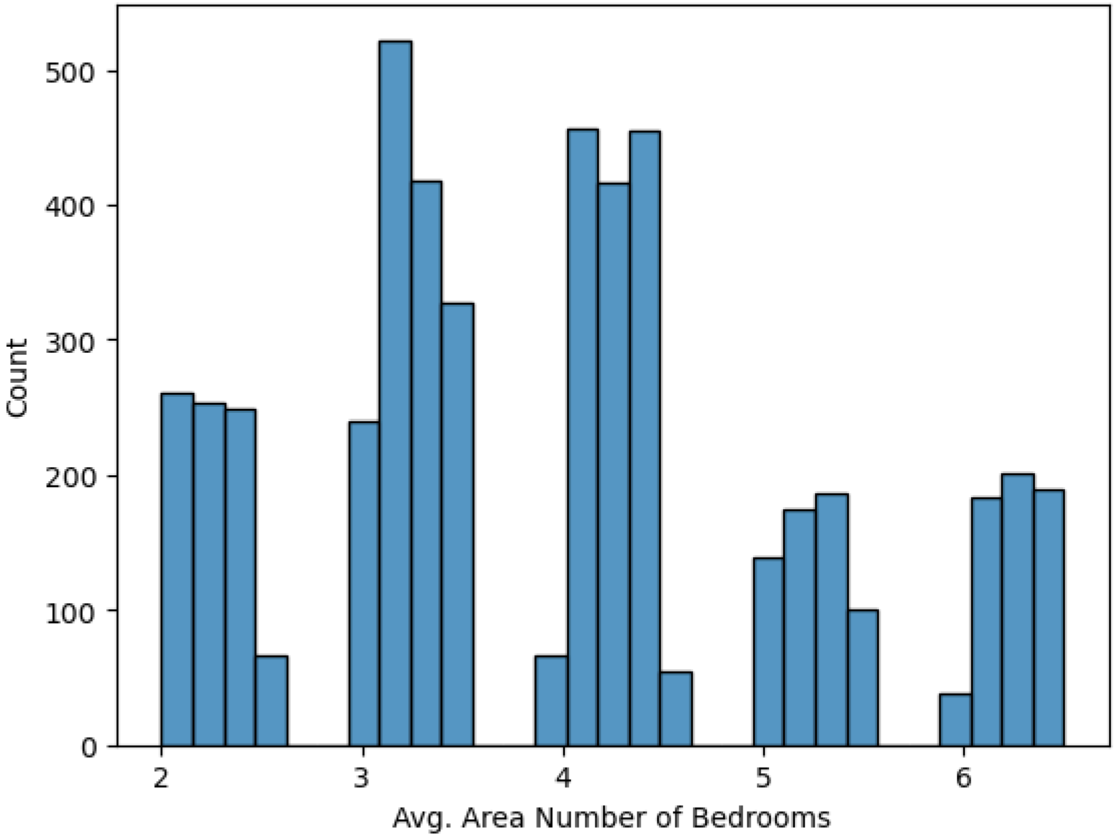


In [18]:
```
sns.histplot(INDIAhousing['Avg. Area Number of Rooms'])
```
Out[18]: <Axes: xlabel='Avg. Area Number of Rooms', ylabel='Count'>



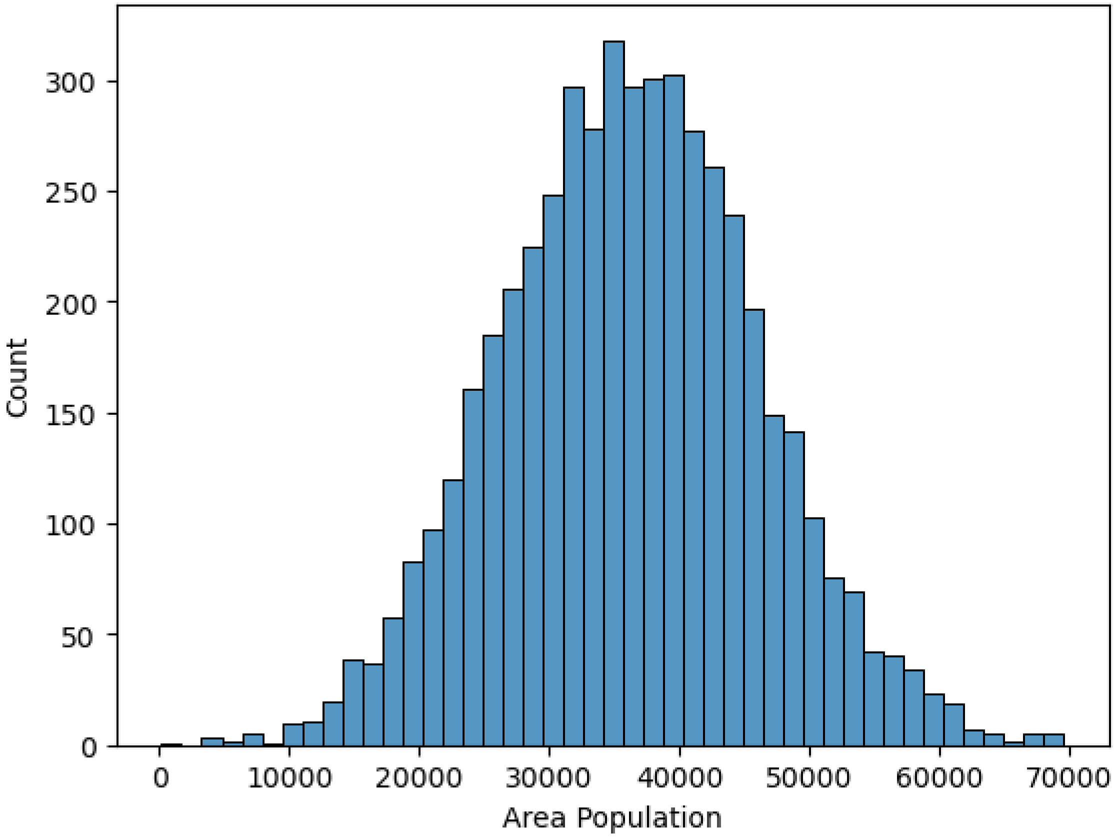
In [19]:
```
sns.histplot(INDIAhousing['Avg. Area Number of Bedrooms'])
```

Out[19]: <Axes: xlabel='Avg. Area Number of Bedrooms', ylabel='Count'>



In [20]:
```
sns.histplot(INDIAhousing['Area Population'])
```

Out[20]: <Axes: xlabel='Area Population', ylabel='Count'>



In [21]:
```
INDIAhousing_numeric = INDIAhousing.drop(columns=['Address'])
```

In [22]:
```
sns.heatmap(INDIAhousing_numeric.corr(), annot=True)
```

Out[22]:
```
<Axes: >
```

Training a Linear Regression Model

In [23]:
```
X = INDIAhousing[['Avg. Area Income','Avg. Area House Age','Avg. Area Number of Roc

Y = INDIAhousing['Price']
```

Split Data into Train ,Test

In [24]:
```
from sklearn.model_selection import train_test_split
```

In [25]:
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.4, random_sta
```

Creating and Training the Linear Regression Model

In [26]:
```
from sklearn.linear_model import LinearRegression
```

In [27]:
```
lm = LinearRegression()
```

In [28]:
```
lm.fit(X_train,Y_train)
```

Out[28]:
```
▾ LinearRegression
LinearRegression()
```

Linear Regression Model Evaluation

In [29]:
```
print(lm.intercept_)
```

```
-2640159.79685267
```

In [30]:
```
coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

Out[30]:

	Coefficient
Avg. Area Income	21.528276
Avg. Area House Age	164883.282027
Avg. Area Number of Rooms	122368.678027
Avg. Area Number of Bedrooms	2233.801864
Area Population	15.150420

Interpreting the coefficients:

Holding all other features fixed, a 1 unit increase in Avg. Area Income is associated with an **increase of $21.52**

Holding all other features fixed, a 1 unit increase in Avg. Area House Age is associated with an **increase of $164883.28**

Holding all other features fixed, a 1 unit increase in Avg. Area Number of Rooms is associated with an **increase of $122368.67**

Holding all other features fixed, a 1 unit increase in Avg. Area Number of Bedrooms is associated with an **increase of $2233.80**

Holding all other features fixed, a 1 unit increase in Area Population is associated with an **increase of $15.15**

Does this make sense? Probably not because I made up this data. If you want real data to repeat this sort of analysis, check out the boston dataset:

from sklearn.datasets import load_boston

boston = load_boston()

print(boston.DESCR)

boston_df = boston.data

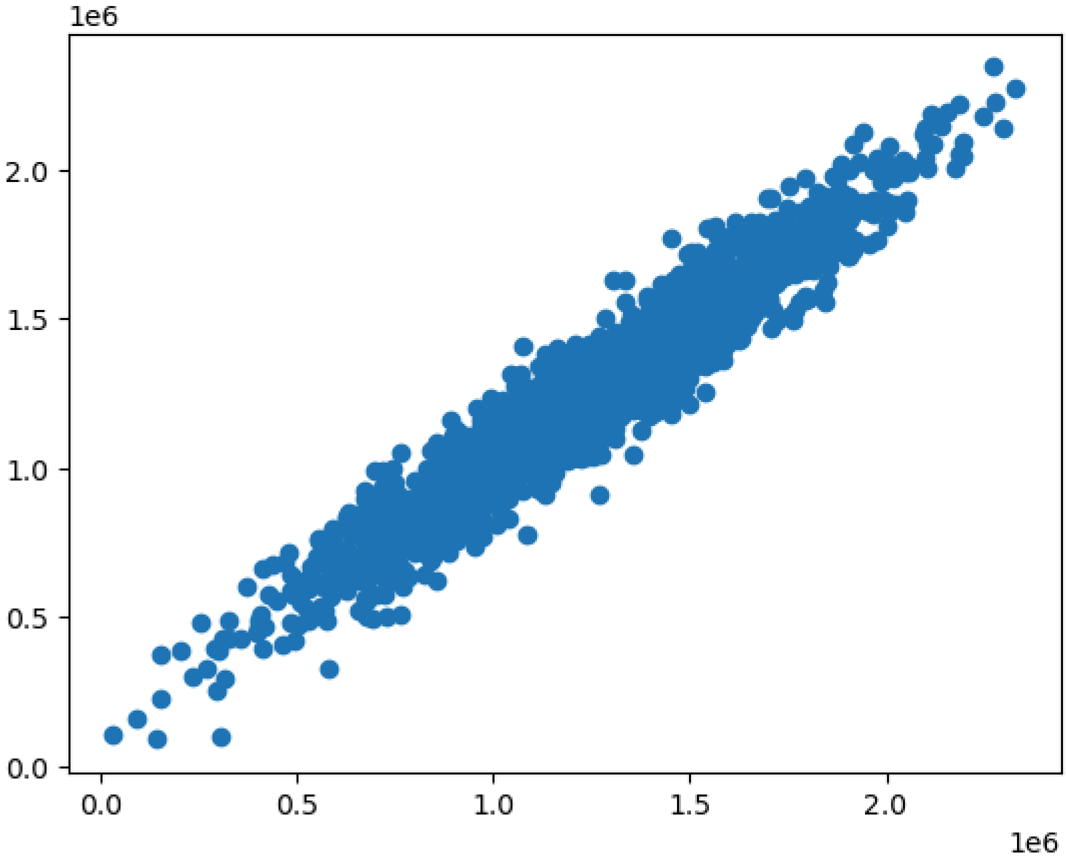
Predictions from our Model

Let's grab predictions off our test set and see how well it did!

In [31]:
```
predictions = lm.predict(X_test)
```

In [32]:
```
plt.scatter(Y_test,predictions)
```

Out[32]:
```
<matplotlib.collections.PathCollection at 0x2138b335750>
```



Residual Histogram

In [33]:
```
sns.distplot((Y_test-predictions),bins=50);
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_4772\1960946261.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

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  sns.distplot((Y_test-predictions),bins=50);
```
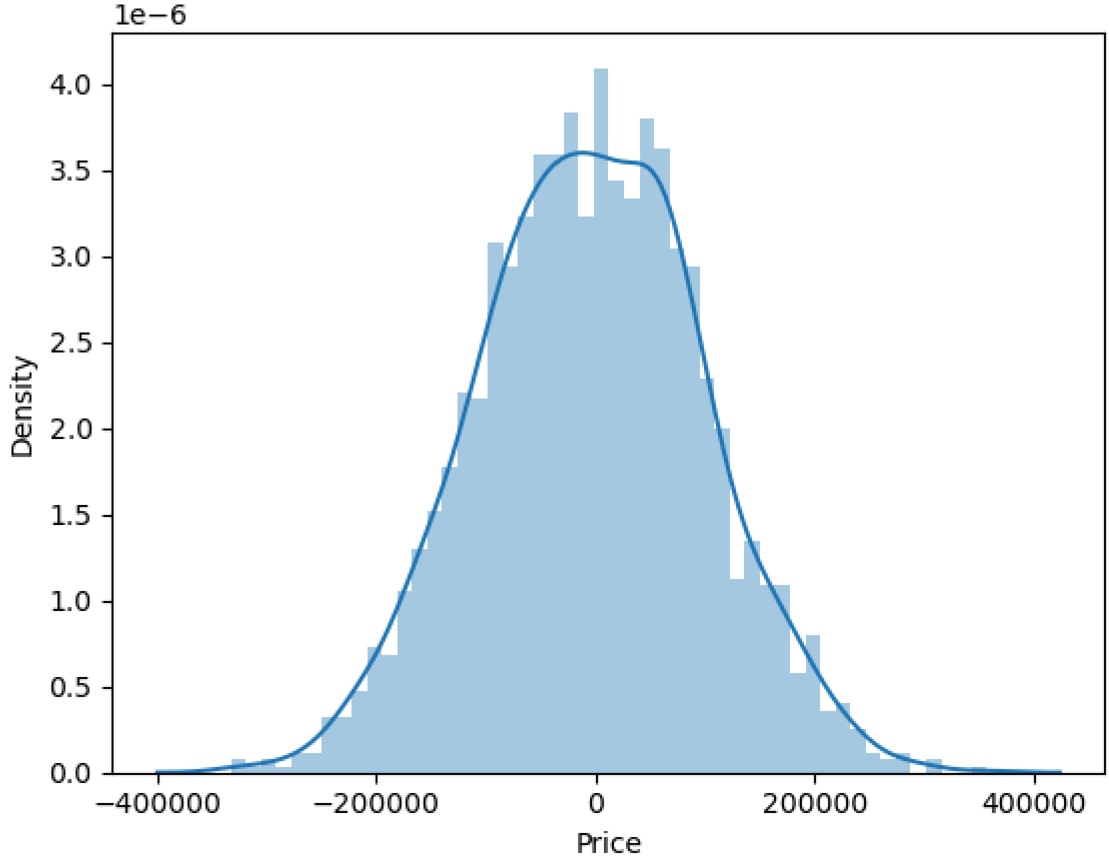
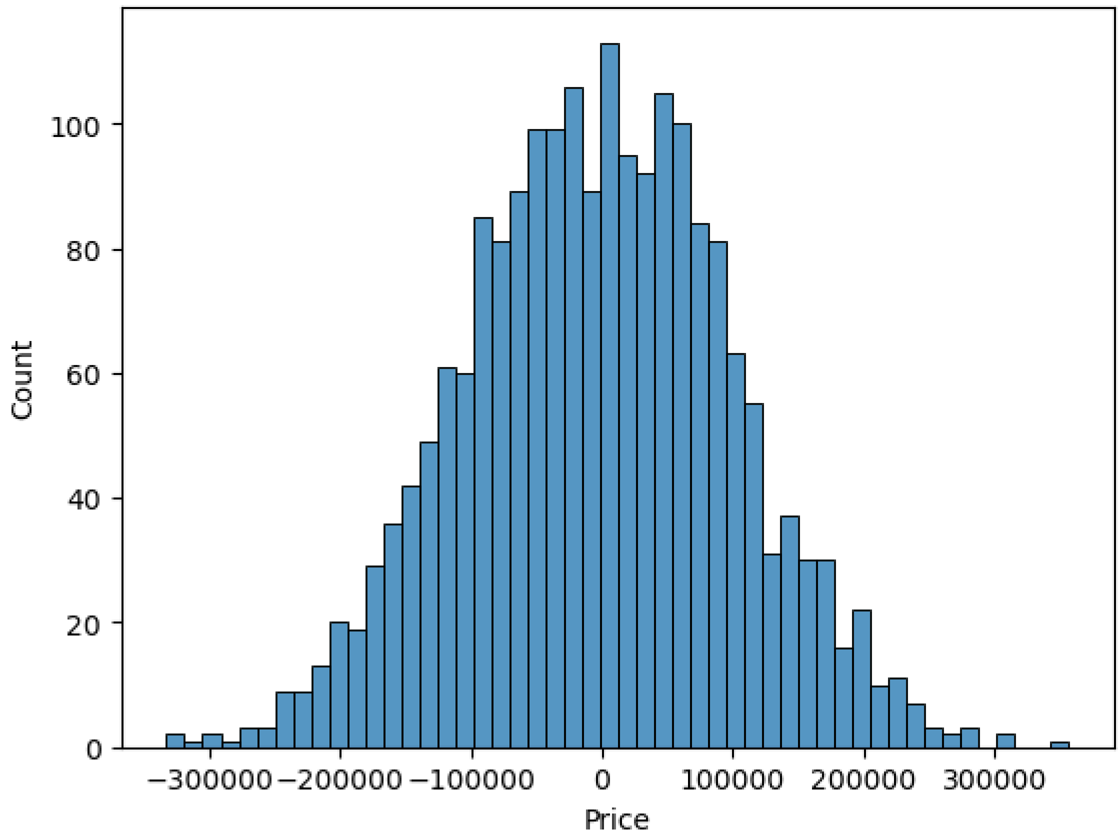



In [34]:
```
sns.histplot((Y_test-predictions),bins=50);
```



Regression Evaluation Metrics

In [35]:
```
from sklearn import metrics
```

In [36]:
```
print('MAE:', metrics.mean_absolute_error(Y_test, predictions))
print('MSE:', metrics.mean_absolute_error(Y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(Y_test, predictions)))
```

```
MAE: 82288.22251914942
MSE: 82288.22251914942
RMSE: 102278.82922290884
```

Thank You

In [ ]:
```

```